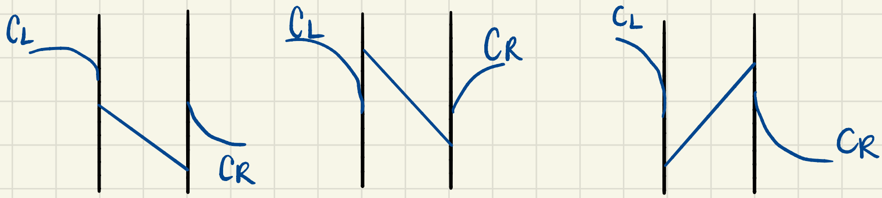
Homework 11

Jenny Tello

1.



$C_L > C_R$

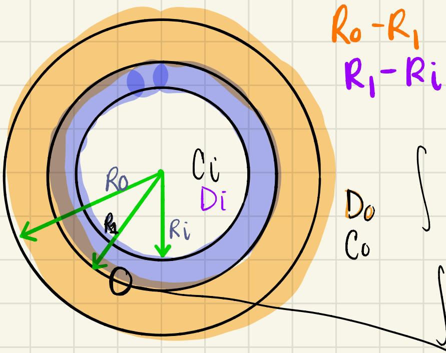
$\phi = C_m/C_L$ or C_m/C_R

$\phi = \frac{C_{membrane}}{C_{medium}}$

	(a)	(b)	(c)
$\frac{C_m}{C_L}$	$\phi < 1$	$\phi > 1$	$\phi < 1$
$\frac{C_m}{C_R}$	$\phi < 1$	$\phi < 1$	$\phi > 1$
	This concentration profile makes sense since the conc. starts of high and then goes down. –possible	This concentration profile doesn't really work. C_R is shown as increasing but it should be decreasing –not possible	This concentration profile also does not work. $\frac{dm}{dx} = \frac{C_1 - C_2}{\ell} =$ but in this case, the slope is positive thus, meaning the concentra. is increasing in the membrane –not possible.

50

② Consider a two layer model of artery. The layers are of thickness: $R_o - R_1$ and $R_1 - R_i$. The inner layer has a diffusion coefficient D_i and the outer layer has the diffusion coefficient D_o. The solute concentration in the lumen $(0 < r < R_i)$ is C_i, and the concentration at R_o is C_o. Calculate the effective diffusion coefficient.



Fick's Law: $\cancel{\frac{dc}{dx}}^{SS} = D\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{dc}{dr}\right)$

$$\int 0 = \int D\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{dc}{dr}\right)$$

$$\int A = \int \cancel{\tfrac{1}{r}}\frac{\partial}{\partial r}\left(\cancel{r}\frac{dc}{dr}\right) \Rightarrow A = r\frac{dc}{dr}$$

$$\int \frac{A}{r} = \int \frac{dc}{dr} \Rightarrow \int \frac{A}{r}\,dr = \int dc$$

$$A\ln(r) + B = C$$

① $C_1(r) = A_1\ln(r) + B_1$

② $C_2(r) = A_2\ln(r) + B_2$

Boundary Conditions: (4)

① @ $C_1(R_i) = C_i$

② @ $C_2(R_o) = C_o$

③ $C_1(R_1) = C_2(R_1)$

④ $-D_1\frac{dC_1}{dr}\Big|_{R_1} = -D_2\frac{dC_2}{dr}\Big|_{R_1}$

① $B_1 = C_i - A_1\ln(R_i)$

② $B_2 = C_o - A_2\ln(R_o)$

③ $C_i + A_1\ln(R_1/R_i) = C_o + A_2\ln(R_1/R_o)$

$$A_1 = \frac{C_o + A_2\ln(R_1/R_o) - C_i}{\ln(R_1/R_i)} \rightarrow \frac{C_o - C_i + A_2\ln(R_1/R_o)}{\ln(R_1/R_i)}$$

④ $$\frac{C_o - C_i + A_2\ln(R_1/R_o)}{\ln(R_1/R_i)}\left(\frac{D_1}{R_1}\right) = \frac{D_2A_2}{R_1}$$

$$\frac{R_1(C_o - C_i + A_2\ln(R_1/R_o))}{\ln(R_1/R_i)\,D_2}\left(\frac{D_1}{R_1}\right) = A_2$$

...

$$A_1 = \frac{D_2(C_0 - C_1)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_0)} \qquad A_2 = \frac{D_1(C_0 - C_1)}{D_2 \ln(R_1/R_2) - D_1 \ln(R_1/R_0)}$$

$$\therefore \quad C_1(r) = C_i + \frac{D_2(C_0 - C_1)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_0)} (\ln r/R_i)$$

$$C_2(r) = C_0 + \frac{D_1(C_0 - C_1)}{D_2 \ln(R_1/R_2) - D_1 \ln(R_1/R_0)} \ln(r/R_0)$$

45